

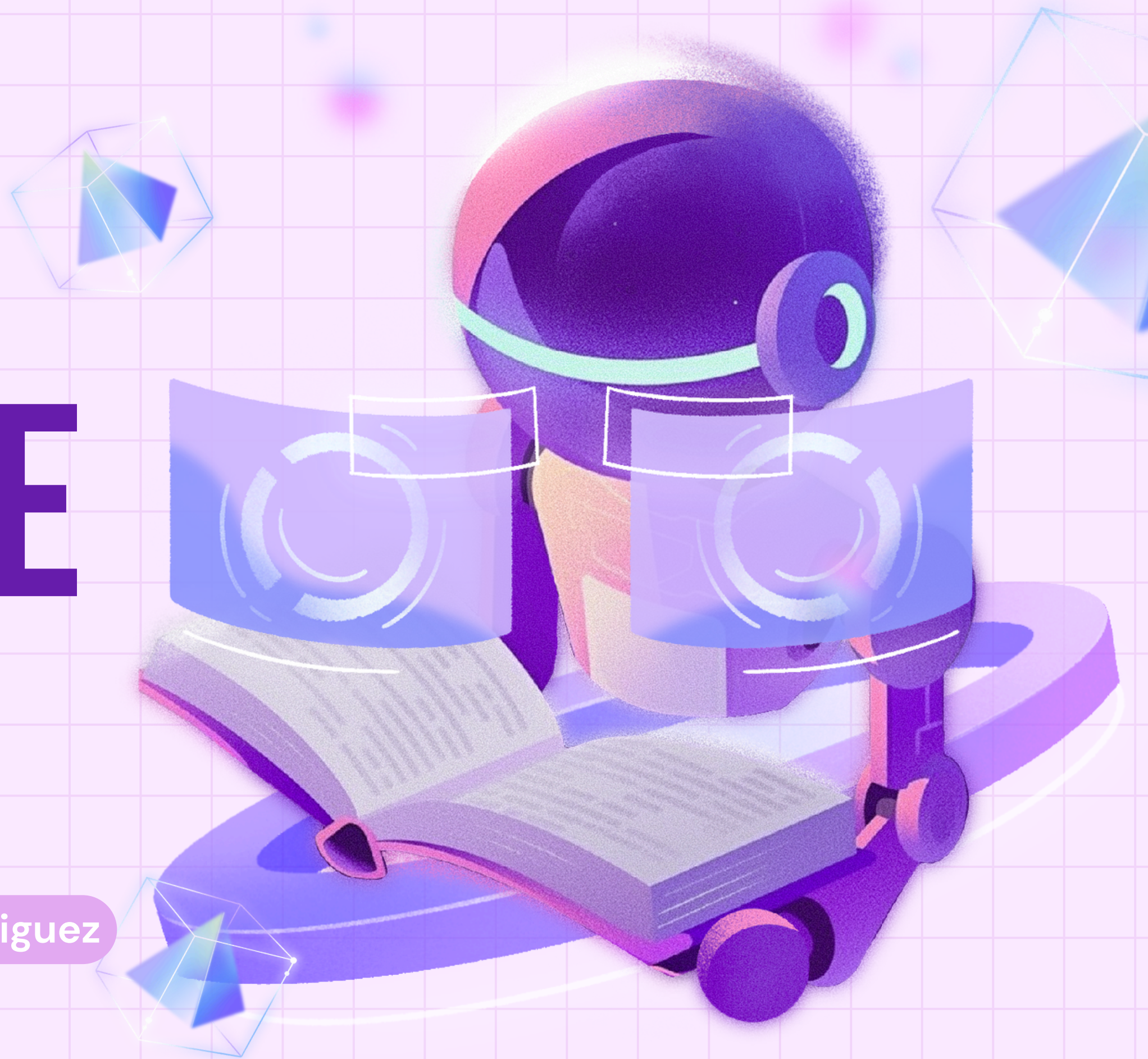
HOUSTON COMMUNITY COLLEGE

ARTIFICIAL INTELLIGENCE

ITAI 1371 FINAL EXAM

May 7, 2026

Presented by Michael Rodriguez



PROJECT OVERVIEW



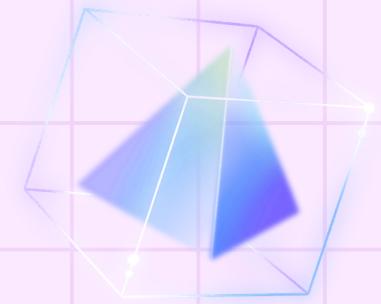
The goal of this project was to use machine learning models to predict whether a borrower is likely to default on a loan. Multiple classification models were trained and compared to determine which model performed best for credit risk prediction.

DATA PREPARATION

AND MACHINE LEARNING MODELS

Data Preparation

- Handled missing values
- Removed unrealistic outliers
- Encoded categorical variables
- Applied MinMax Scaling
- Created debt-to-income ratio feature

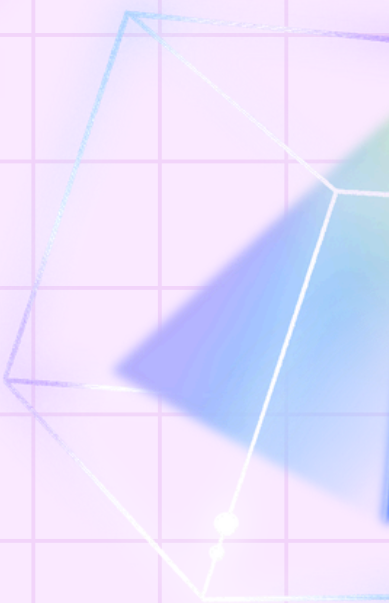


Data Split

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

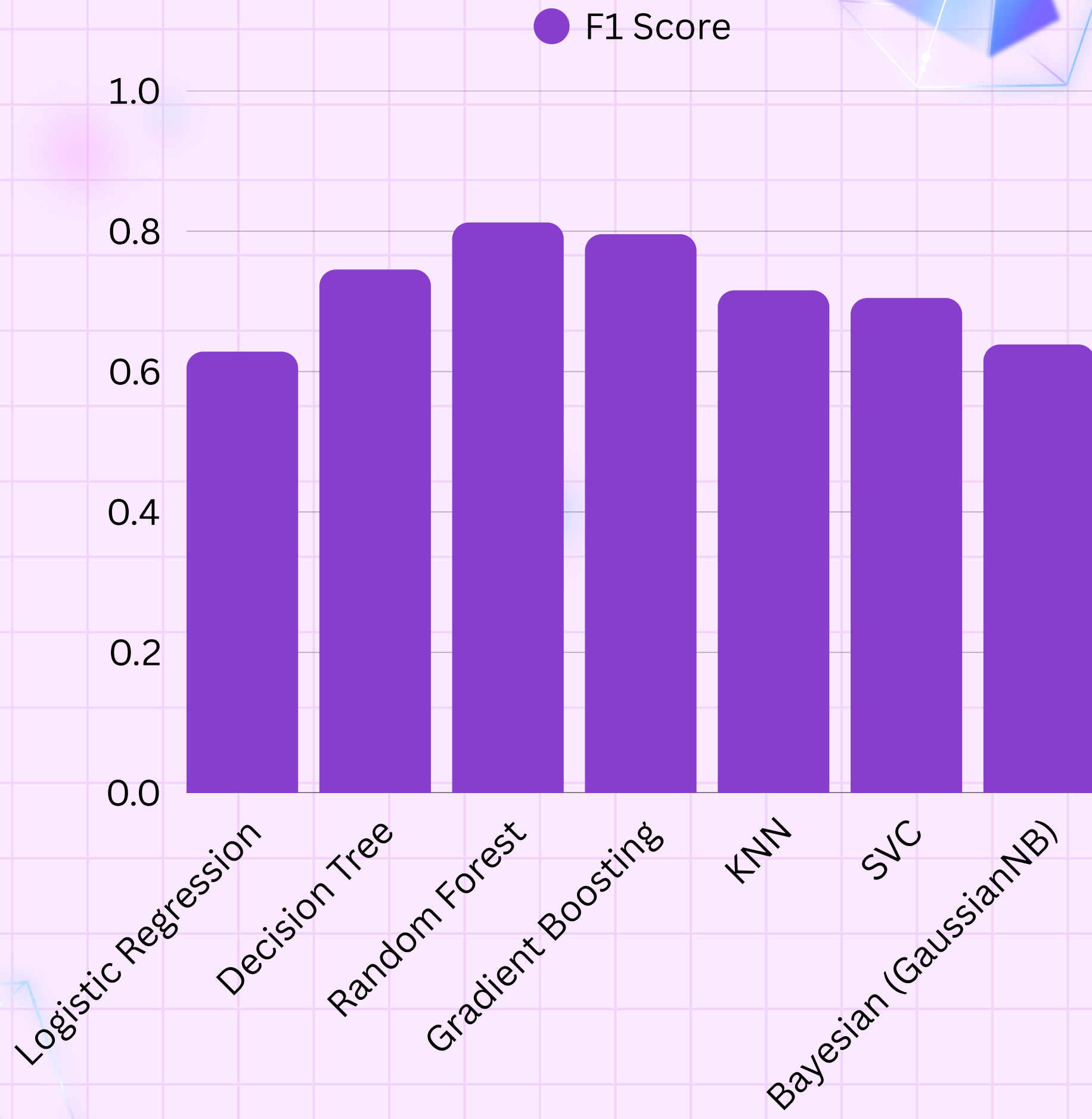
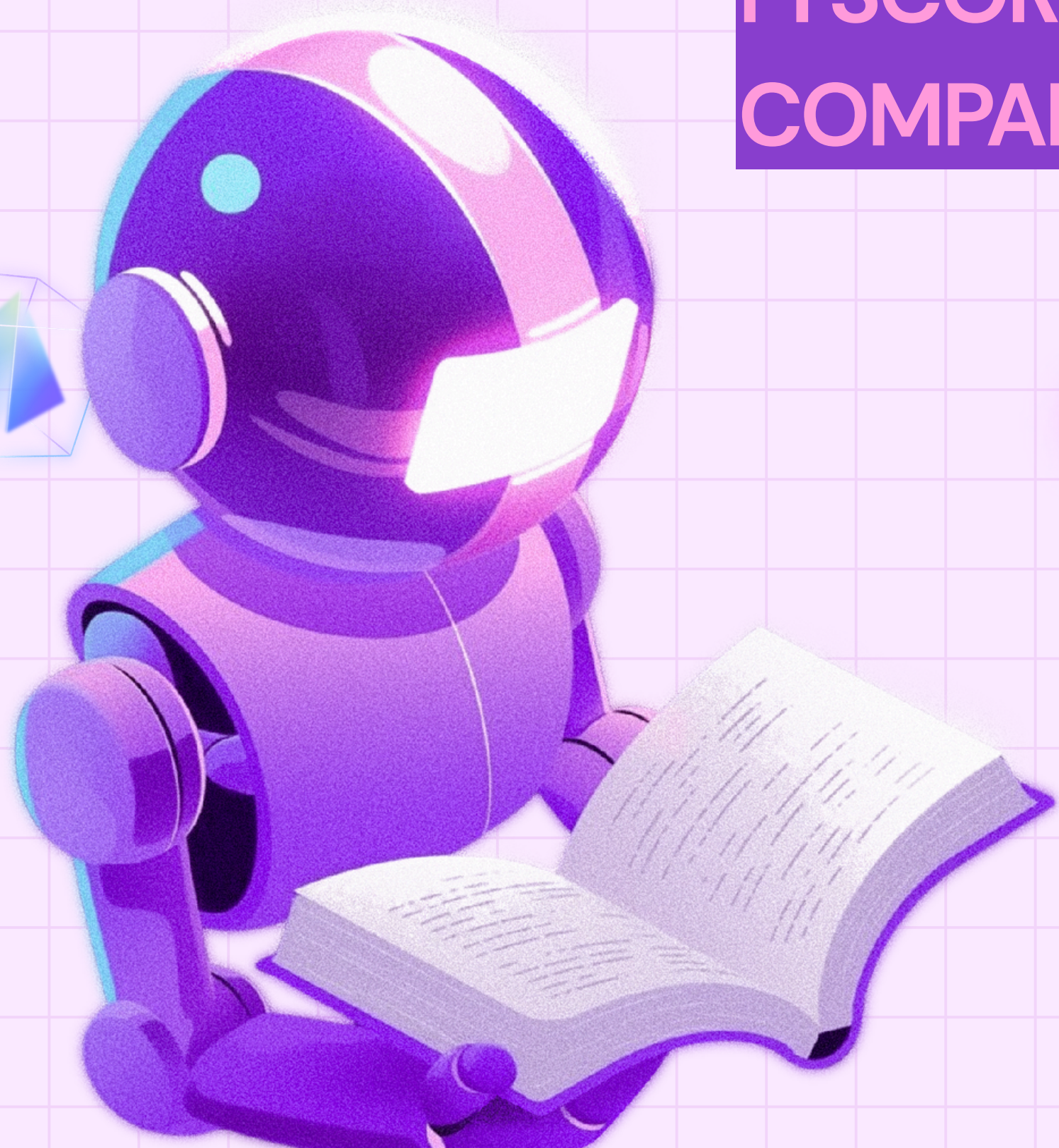
Models Trained

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- KNN
- SVC
- Gaussian Naive Bayes



MODEL RESULTS

F1 SCORE COMPARISON



ENSEMBLE MODEL

TESTING RESULTS



Ensemble Model

- Random Forest
- Gradient Boosting
- SVC

Ensemble Test Results

- Accuracy: 93.2%
- F1 Score: 0.822
- ROC-AUC: 0.932

The ensemble model improved prediction stability and produced strong overall performance for credit risk classification!



THANK YOU

FOR YOUR ATTENTION!